Diploma in Information Technology

信息技术大专文凭

Business Statistic with Python

商务统计（Python应用）

Instructions for CA2 Group Assignment

CA2小组作业说明

April 2026 Term

2026年4月学期

Continuous Assessment 2 - Group Assignment (40%)

持续评估2 - 小组作业（占总成绩40%）

The objective of this assignment is for students to apply statistical concepts on real-world data and write a report on their observation. Students must ensure that their observation is supported by valid sources.
本作业旨在让学生将统计概念应用于真实世界数据，并就观察结果撰写报告。学生须确保观察结论有可靠来源支撑。

This assignment will make use of the what you have learnt in:
本作业将运用你在以下课程内容中所学的知识：

- Organising and visualising variables
  变量的整理与可视化
- Numerical Descriptive Measures
  数值描述性指标
- Use of Statistical Software for Data Analysis (Microsoft Excel)
  统计软件在数据分析中的应用（微软Excel）
- Hypothesis Tests
  假设检验
- Simple Linear Regression and Correlation Coefficient
  简单线性回归与相关系数
- Multiple Linear Regression
  多元线性回归

Task 1

任务1

Over the past decade, Singapore has emerged as one of Asia's leading smart cities, driven by strong government support for digital transformation, data-driven decision-making, and technology innovation.
过去十年，在政府大力支持数字化转型、数据驱动决策与技术创新的推动下，新加坡已跻身亚洲领先智慧城市之列。

One of the sectors that has undergone rapid change is retail, particularly within large shopping malls and commercial districts such as Orchard Road, Marina Bay, and suburban town centres.
零售业是经历快速变革的行业之一，尤其是在乌节路、滨海湾及郊区城镇中心等大型购物中心与商业区。

Modern retail environments are no longer limited to physical shelves and cashiers.
现代零售场景不再局限于实体货架与收银台。

Instead, they increasingly rely on smart technologies such as Internet of Things (IoT) sensors, artificial intelligence (AI), mobile payment systems, and customer analytics platforms.
相反，其愈发依赖物联网（IoT）传感器、人工智能（AI）、移动支付系统及客户分析平台等智能技术。

These technologies enable retailers to monitor foot traffic, analyse customer behaviour, manage inventory in real time, and offer personalised promotions.
这些技术助力零售商监测客流量、分析客户行为、实时管理库存，并推出个性化促销活动。

For example, cameras and sensors can track how shoppers move within a store, while digital payment platforms and loyalty apps generate valuable data on purchasing patterns.
例如，摄像头与传感器可追踪顾客在店内的动线，数字支付平台与会员应用则能生成关于消费模式的宝贵数据。

In Singapore, these developments are strongly aligned with the Smart Nation initiative, which encourages businesses to use digital technologies to improve productivity, customer experience, and sustainability.
在新加坡，这些发展与"智慧国度"计划高度契合，该计划鼓励企业运用数字技术提升生产力、优化客户体验并增强可持续性。

Large retailers such as FairPrice, Sheng Siong, and shopping mall operators have already begun experimenting with smart checkout systems, automated inventory tracking, and AI-based customer engagement tools.
职总平价、昇菘等大型零售商及购物中心运营商已开始试点智能收银系统、自动化库存追踪及基于人工智能的客户互动工具。

As these technologies become more affordable and widespread, smart retail is expected to play an even more important role in shaping the future of shopping in Singapore.
随着这些技术成本降低、普及度提升，智慧零售预计将在塑造新加坡购物未来的过程中发挥更为重要的作用。

DELIVERABLE – TASK 1

任务1 交付成果

You are required to conduct online research on smart retail technologies in Singapore. Using one or more credible online sources, you will analyse how digital and smart technologies are transforming physical retail environments.
你需针对新加坡智慧零售技术开展线上调研，借助一个或多个可靠线上来源，分析数字与智能技术如何重塑实体零售场景。

You should discuss three possible future applications of smart retail technologies in Singapore that could help to:
你需论述新加坡智慧零售技术未来三种可行的应用方向，这些应用可助力实现：

- Improve the shopping experience for customers
  提升顾客购物体验

- Increase business efficiency and profitability for retailers
  提高零售商的运营效率与盈利能力
- Support better urban and commercial planning
  支撑更完善的城市与商业规划

Your report should not be longer than 2 pages.
报告篇幅不得超过2页。
(10 marks)
（10分）

Task 2

任务2

Refer to retail_store_data.csv
请参考`retail_store_data.csv`文件

Variable	Description
store_size_sqm	Total floor area of the store (square metres)
monthly_rent_sgd	Monthly rental cost (SGD)
foot_traffic	Average number of visitors per day
staff_count	Number of employees working in the store
promotion_days	Number of promotional days per month
store_type	Type of store (Fashion, Electronics, Food)
monthly_sales_sgd	Monthly sales revenue (SGD)

Problem Description:

问题描述

You have recently joined UrbanMall Analytics Pte Ltd as a Junior Data Analyst.
你近期以初级数据分析师的身份加入城市商场分析私人有限公司。

UrbanMall Analytics Pte Ltd is a Singapore-based data consultancy that works with major shopping mall operators, retail chains, and urban planners to improve the performance of physical retail spaces using data-driven insights.
城市商场分析私人有限公司是一家总部位于新加坡的数据咨询公司，与大型购物中心运营商、零售连锁企业及城市规划师合作，通过数据驱动洞察提升实体零售空间的运营表现。

With increasing competition from e-commerce platforms and rising operating costs, shopping malls and retailers must rely on data analytics, customer behaviour tracking, and performance modelling to remain competitive.
面对电商平台日益激烈的竞争与不断攀升的运营成本，购物中心与零售商必须依托数据分析、客户行为追踪与运营建模来保持竞争力。

Modern shopping malls in Singapore are equipped with smart sensors, Wi-Fi tracking systems, point-of-sale (POS) platforms, and loyalty programmes that collect detailed information about how customers interact with retail stores.
新加坡的现代化购物中心配备了智能传感器、Wi-Fi追踪系统、销售点（POS）平台及会员计划，可收集顾客与零售店铺互动的详细信息。

These systems generate large volumes of data, including information on foot traffic, store size, promotional activity, staffing levels, and sales revenue.
这些系统会生成海量数据，涵盖客流量、店铺面积、促销活动、员工配置及销售收入等信息。

When properly analysed, this data can help mall managers and retailers answer important questions such as:
经合理分析后，这些数据可帮助商场管理者与零售商解答以下关键问题：

- Which types of stores perform better in high-traffic locations?
  哪些类型的店铺在高客流量地段表现更优？
- How does store size and rental cost affect profitability?
  店铺面积与租金成本如何影响盈利能力？
- Do promotional activities significantly increase sales?
  促销活动是否能显著提升销售额？
- What combination of factors best predicts a store's monthly revenue?
  哪些因素组合能最精准地预测店铺月收入？

To support strategic decision-making, UrbanMall Analytics has collected operational and financial data from 600 retail stores located across multiple shopping malls in Singapore.
为支撑战略决策，城市商场分析公司收集了新加坡多家购物中心内600家零售店铺的运营与财务数据。

These stores represent a mixture of fashion, electronics, and food outlets, providing a realistic cross-section of Singapore's retail landscape.
这些店铺涵盖服装、电子、食品品类，真实反映了新加坡的零售行业格局。

You have been assigned, as a Junior Data Analyst, to analyse this dataset and produce meaningful insights that can be used by mall operators and retail executives to guide investment, leasing decisions, and marketing strategies.
作为初级数据分析师，你需对该数据集展开分析，提炼有价值的洞察，为商场运营商与零售高管的投资、租赁决策及营销策略提供指导。

Using the tools you have learned in Business Statistics with Python (BSP), you need to perform exploratory data analysis, apply statistical methods, and build predictive models to evaluate and understand the trends and relationships hidden in the data.
你需运用商务统计（Python应用）课程所学工具，开展探索性数据分析，应用统计方法并构建预测模型，挖掘并理解数据中隐藏的趋势与关联。

(a) Read through the core module "Data Analytics" before you attempt this part of the question. Data preprocessing is a crucial step in data analysis. If the data itself contains typo, missing values and outliers etc, it can significantly impact the results of data analysis. Please discuss the importance of data preprocessing and two aspects of data preprocessing. If you feel you need to preprocess the data, please do so for further analysis. If not, justify your decision.
(a) 作答本部分前，请通读"数据分析"核心模块内容。数据预处理是数据分析的关键步骤，若数据本身存在拼写错误、缺失值、异常值等问题，会严重影响数据分析结果。请论述数据预处理的重要性及数据预处理的两个方面。若你认为需要对数据进行预处理，请完成预处理以便后续分析；若无需预处理，请说明理由。

(10 marks)
（10分）

(b) By using Excel, compute key summary statistics (central tendency and spread) of monthly_sales_sgd, grouped by store_type. Tabulate and compare the results and comment on the differences in sales patterns among Fashion, Electronics, and Food stores. Your comments should focus on the data spread, locality and shape of the data.
(b) 使用Excel计算按店铺类型分组的月销售收入（monthly_sales_sgd）的核心描述性统计量（集中趋势与离散程度）。将结果制表并对比，评述服装、电子、食品类店铺的销售模式差异，评述需聚焦数据的离散程度、集中位置与分布形态。
(5 marks)
（5分）

(c) Draw a histogram of the attribute “monthly_sales_sgd”. Interpret the graphs.
(c) 绘制月销售收入（monthly_sales_sgd）的直方图并解读图表。
(6 marks)
（6分）

(d) Construct and interpret a 95% confidence interval around the monthly rental cost of stores with store_size_sqm > 100. Clearly indicate any assumptions made.
(d) 为店铺面积大于100平方米的店铺的月租金构建95%置信区间并解读，明确说明所做的假设。
(4 marks)
（4分）

(e) A media report claims that if there is a relationship between the “store_type” and “promotion intensity”. Do you agree? Perform a Chi-square test at the 5% significance level. Indicate any assumptions made.
(e) 某媒体报道称店铺类型与促销力度之间存在关联。你是否认同该观点？在5%显著性水平下开展卡方检验，并说明所做假设。
(10 marks)
（10分）

(f) Among all attributes, which 3 do you think are most relevant to predicting “monthly_sales_sgd”? Please explain your reason.
(f) 在所有变量中，你认为哪三个变量与预测月销售收入（monthly_sales_sgd）关联性最强？请说明理由。
(10 marks)
（10分）

(g) Construct a multiple linear regression model and predict the “monthly sales revenue”. If you are given a new record about a installation (Store size = 95 sqm, Rent = $15,000, Foot traffic = 500 customers/day, Staff = 7, Promotions = 8 days, Store type = Electronics), what is the predicted result of the “monthly sales revenue”?
(g) 构建多元线性回归模型并预测月销售收入。现有一条新店铺数据：店铺面积95平方米、租金15000新加坡元、日均客流量500人、员工7人、月促销8天、店铺类型为电子类，其月销售收入预测值为多少？
(10 marks)
（10分）

(h) Lastly, you need to present your key findings, in Part 3, of your analysis to the Manager.
(h) 最后，你需在第三部分向经理汇报分析的核心结论。
(10 marks)
（10分）

Part 1 Excel Analysis

第一部分 Excel分析

This part of your assignment will require the use of Microsoft Excel. Your working steps should be carefully organized using worksheet for each question supporting your reports in Part 2.
本部分作业需使用微软Excel完成。请为每个问题单独创建工作表，清晰梳理操作步骤，为第二部分的报告提供支撑。

Part 2 Professional Report

第二部分 专业报告

Write a report answering the above questions, in sequence. You may need to snip/crop certain relevant results/output from your Excel workbook and incorporate them into your report. The Report needs to be standalone. Do NOT make reference to the Excel workbook within the Professional report.
按顺序撰写报告解答上述问题，可截取Excel工作簿中的相关结果/输出图表插入报告。报告需独立成文，**不得**在专业报告中提及Excel工作簿。

Your written professional report should comply with the following presentation standards:
撰写的专业报告需符合以下排版规范：

1. Typed using a standard professional font type (e.g. Times Roman), 12-point font size.
    1. 使用标准专业字体（如Times New Roman），字号12磅。
2. 1.5-line spacing, numbered pages, and clear use of titles and section headings.
    2. 1.5倍行距，标注页码，清晰使用标题与分节标题。
3. Checked for spelling, typographical and grammatical errors.
    3. 检查拼写、排版与语法错误。
4. Where relevant, round to 3 decimal places, except for currency (e.g. $35.56) and percentage (e.g. 26.75%).
    4. 相关数值保留三位小数，货币（如35.56新加坡元）与百分比（如26.75%）除外。
5. With all relevant tables and charts, the report should be no more than 10 pages long, excluding cover page, reference list.
    5. 包含所有相关表格与图表后，报告篇幅不超过10页（封面、参考文献列表除外）。

Part 3 Presentation

第三部分 汇报展示

This portion comprises of 2 components: (a) ppt slides (b) presentation. Oral Presentation based on Power point slides (max=20 slides). Perform an oral presentation (max = 15 mins) of your answers in Task 2, Part 2. Briefly explaining the highlight of your analysis, findings and conclusions.
本部分包含两项内容：(a) PPT幻灯片 (b) 口头汇报。基于PPT幻灯片进行口头汇报（幻灯片最多20页），针对任务2第二部分的答案进行时长不超过15分钟的口头展示，简要说明分析重点、研究发现与结论。
(25 marks)
（25分）

IMPORTANT NOTE

重要提示

1. Deadline is 11th May 2026, 11:59 am
    1. 截止时间：2026年5月11日上午11:59
2. Multiple submissions are allowed; only latest submission will be marked. You are required to submit/upload the following documents onto Canvas in the following sequence:
    2. 允许多次提交，仅评分最新一次提交的内容。请按以下顺序在Canvas平台提交/上传以下文件：
       a) Excel Analysis - *.xlsx => Task 2 Part 1
       a) Excel分析文件 - *.xlsx => 任务2第一部分
       b) Professional Report - *. docx => Task 1 + Task 2 Part 2
       b) 专业报告 - *.docx => 任务1+任务2第二部分
       c) Presentation - *.pptx => Task 2 Part 3
       c) 汇报展示文件 - *.pptx => 任务2第三部分
3. The structure of your MS Word report must include:
    3. Word报告的结构必须包含：
    1. Cover Page comprising of:
        1. 封面，需包含：
           a) Institution name (and institution logo) of this programme
           a) 本课程所属院校名称（及院校校徽）
           b) Module name, semester and year
           b) 课程名称、学期与年份
           c) Date of submission
           c) 提交日期
           d) Student’s name and student ID number
           d) 学生姓名与学号
    2. Task 1 report
        2. 任务1报告
    3. Task 2 (Part 2) report
        3. 任务2（第二部分）报告
           a) An Executive Summary,
           a) 执行摘要
           b) An Introduction,
           b) 引言
           c) Analysis by questions,
           c) 分问题分析
           d) Conclusions.
           d) 结论
    4. Reference List
        4. 参考文献列表
    5. The references should be presented in Harvard format and should have at least THREE (3) references.

5. 参考文献需采用哈佛格式，且至少包含3篇参考文献。

Report Word Limit

报告字数限制

Combined Task 1 and Task 2: Max = 1500 to 2000 words
任务1与任务2合计：最多1500-2000单词

Penalty Marks for Late Submission

迟交扣分规则

By one day: 20% to be deducted from total marks.
迟交1天：总分扣20%
More than one day: submission will NOT be graded.
迟交超过1天：作业不予评分

Lecturer Contact

讲师联系方式

You should contact your lecturer via your SIM email whenever you have any issue about your project.
若对作业有任何疑问，请通过SIM邮箱联系课程讲师。

评分标准（Scoring Criteria）

S/N	Criteria	Excellent	Very good	Good	Acceptable	Weak	Weak
1	Task 1 (10%)	Student has a clear understanding of the question requirements and is able to produce clear and accurate description supported by summary statistics and relevant charts effectively.	Student has a clear understanding of the question requirements and is able to produce clear description supported by summary statistics and relevant charts effectively.	Student has a clear understanding of the question requirements and is able to produce suitable description supported by some government's statistics and charts.	Student has a reasonable understanding of the question requirements and can produce description supported by a few government's statistics data and charts.	Student do not have a clear understanding of the question Requirements. Lack of clear description. Do not have sufficient summary statistics to support answers.	

S/N	Criteria	Excellent	Very good	Good	Acceptable	Weak	Weak
2	Task 2 (65%)	Student has a clear understanding of the question requirements and can provide answers with clear and coherent explanations and comments.	Student has a clear understanding of the question requirements and can provide most of the answers with clear and coherent explanations and comments.	Student has a clear understanding of the question requirements and can provide correct answers. Explanation could be improved with better and clearer working steps and comments.	Student has an acceptable level of understanding of the question requirements and can provide correct Answers for most of the questions. Explanation could be improved with better and clearer working steps and comments.	Student do not have a clear understanding for most of the question requirements. Lack of clear description.	
3	Oral presentation (25%)	An excellent presentation with clear flow and strong command of oral English coupled with well-illustrated content and rich slides. Could handle questions very well.	A presentation with good oral English and sufficient content on the slides.	A reasonable presentation with sufficient content, though some aspects of the oral English could improve.	Student presented in much less time than allocated; poorly organized and illustrated slides. Oral English could improve.	Student did not present.	

中文对照

序号	评分项	优秀	良好	合格	尚可	差	差
1	任务1（10%）	充分理解题目要求，能清晰、准确阐述内容，有效结合统计摘要与相关图表支撑观点	清晰理解题目要求，能清晰阐述内容，有效结合统计摘要与相关图表支撑观点	清晰理解题目要求，能合理阐述内容，结合部分官方统计数据与图表支撑观点	对题目要求有基本理解，能阐述内容，结合少量官方统计数据与图表支撑观点	未清晰理解题目要求，阐述不清晰，无充足统计摘要支撑答案	
2	任务2（65%）	充分理解题目要求，答案准确，解释与评述清晰连贯	清晰理解题目要求，完成大部分作答，解释与评述清晰连贯	清晰理解题目要求，答案正确，可通过更完善的操作步骤与评述优化解释	对题目要求有基本理解，完成大部分题目正确作答，可通过更完善的操作步骤与评述优化解释	对大部分题目要求未清晰理解，阐述不清晰	

序号	评分项	优秀	良好	合格	尚可	差	差
3	口头汇报（25%）	汇报展示优秀，逻辑清晰，英语口语流利，内容阐释充分，幻灯片丰富，能出色应对提问	汇报展示良好，英语口语流畅，幻灯片内容充足	汇报展示合格，内容充足，英语口语可进一步提升	汇报时长远低于规定时长，内容组织混乱，幻灯片效果差，英语口语可提升	未进行汇报展示	